Lab program-12:

12. Design a C program to simulate the concept of Dining-Philosophers problem

C lab_12.c > main()

```c
    // Create mutexes (one for each chopstick)
    for (int i = 0; i < NUM_PHILOSOPHERS; i++)
    {
        chopsticks[i] = CreateMutex(NULL, FALSE, NULL);

        if (chopsticks[i] == NULL)
        {
            printf("Failed to create mutex.\n");
            return 1;
        }
    }

    // Create philosopher threads
    for (int i = 0; i < NUM_PHILOSOPHERS; i++)
    {
        philosopher_ids[i] = i;

        philosophers[i] = CreateThread(
            NULL,
            0,
            philosopherLifeCycle,
            &philosopher_ids[i],
            0,
            NULL
        );

        if (philosophers[i] == NULL)
        {
            printf("Failed to create thread.\n");
            return 1;
        }
    }

    // Wait for all philosopher threads (they run forever)
    WaitForMultipleObjects(
        NUM_PHILOSOPHERS,
        philosophers,
        TRUE,
        INFINITE
    );

    // Cleanup
    for (int i = 0; i < NUM_PHILOSOPHERS; i++)
    {
        CloseHandle(philosophers[i]);
        CloseHandle(chopsticks[i]);
    }

    return 0;
}
```

OUTPUT:

```
PS C:\Users\SMILE\Documents\OS\Lab programs> ./lab_12.exe
Philosopher 3 is thinking...
Philosopher 0 is thinking...
Philosopher 4 is thinking...
Philosopher 2 is thinking...
Philosopher 1 is thinking...
Philosopher 4 is eating...
Philosopher 3 is eating...
```